

ASNT STANDARD

PERSONNEL QUALIFICATION
AND CERTIFICATION FOR IN-LINE
INSPECTION TECHNOLOGIES USED IN
THE EXAMINATION OF PIPELINES

2023
EDITION



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ASNT Mission Statement:
ASNT's mission is to advance the field of nondestructive testing.

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The *ASNT Code of Ethics* was developed to provide members of the Society with broad ethical statements to guide their professional lives. In spirit and in word, each ASNT member is responsible for knowing and adhering to the values and standards set forth in the Society's Code. More information, as well as the complete version of the *Code of Ethics*, can be found on ASNT's website, *asnt.org*.

ANSI/ASNT ILI-PQ-2023
American National Standard
In-line Inspection Personnel Qualification and Certification Standard

The American Society for Nondestructive Testing Inc.
Approved 7 April 2023
American National Standards Institute

Abstract

This standard applies to personnel whose specific tasks or jobs require appropriate knowledge of the technical principles underlying nondestructive testing (NDT) methods for which they have responsibilities within the scope of their employment. These specific tasks or jobs include, but are not limited to, performing, specifying, reviewing, monitoring, supervising, and evaluating NDT work.

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This standard is subject to revision or withdrawal at any time by ASNT.

AMERICAN NATIONAL STANDARD

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Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

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FOREWORD

[This foreword is not a part of American National Standard ANSI/ASNT ILI-PQ-2023]

This standard has been developed by the American Society for Nondestructive Testing Inc., to establish minimum requirements for the qualification and certification of in-line inspection (ILI) personnel whose jobs require specific knowledge of the technical principles of ILI technologies, operations, regulatory requirements, and industry standards as applicable to pipeline systems.

The verb “shall” has been used throughout this document to emphasize the requirements presented herein. It is the employer’s responsibility to address specific needs and to utilize these requirements as appropriate in a written practice.

Inquiries related to this standard should be directed to the ASNT Standards Council at standards@asnt.org. Inquiry forms can be found on the ASNT website (<https://asnt.org/MajorSiteSections/Standards.aspx>).

The In-line Inspection Personnel Qualification and Certification Standard (Document No. ASNT ILI-PQ) was initially drafted, reviewed, and approved by the ASNT ILI Committee, which submitted the standard to the ASNT Standards Council. Committee or council approval of the standard does not necessarily imply that all committee members voted for its approval. At the time it approved this standard, the Standards Council had the following members:

Michael McDaniel, Chair	Gerry Churchwell	Mike Ruddy
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Ethics Statement

The NDT profession plays a critical role in ensuring public safety by providing information on the condition and integrity of critical infrastructure and products through its various NDT methods and techniques. The NDT profession has a responsibility to act ethically in all activities including the proper collection, interpretation, and reporting of NDT data. To this end, ASNT highly encourages all individuals and companies using this standard/document to incorporate the topic of ethics and ethical behavior of its employees and contractors into their training programs and examinations.

SUMMARY OF CHANGES IN THE 2023 EDITION ILI-PQ

The changes made are also indicated with a line in the margin for easy identification.

- | | |
|--|---|
| 1. Various editorial changes made throughout the document including grammar and numbering. | 28. Revised: 6.4. |
| 2. Updated: Section 2.0 Documents revisions/editions. | 29. Added: 6.5. |
| 3. Revised: 3.1 Advance Data Analyst definition. | 30. Revised: 6.7. |
| 4. Revised: 3.2 Advanced Data Analysis definition. | 31. Revised: 6.7.1. |
| 5. Revised: 3.6 Anomaly definition. | 32. Revised: 7.2. |
| 6. Added: 3.8 Appurtenance definition. | 33. Revised: 7.3. |
| 7. Added: 3.9 Axial Magnetic Flux Technology definition. | 34. Revised: 8.1.1. |
| 8. Added: 3.15 Component definition. | 35. Revised: 8.1.2. |
| 9. Added: 3.22 Electromagnetic Acoustic Transducer (EMAT) definition. | 36. Revised: 8.1.4. |
| 10. Revised: 3.23 Employer definition. | 37. Revised: 8.2.1. |
| 11. Revised: 3.25 Feature definition. | 38. Revised: 8.2.2. |
| 12. Revised: 3.32 In-Field Inspection Data Confirmation definition. | 39. Revised: 8.3.3. |
| 13. Added: 3.40 Mapping Pig definition. | 40. Revised: 9.6.1. |
| 14. Revised: 3.48 Report Generation definition. | 41. Revised: 9.6.3. |
| 15. Revised: 3.56 Written Practice definition. | 42. Revised: 12.1. |
| 16. Added: 3.51 Transverse Magnetic Field Technology definition. | 43. Revised: 12.1.1. |
| 17. Added: 3.53 Ultrasonic Compression Wave Technology definition. | 44. Revised: 12.1.2. |
| 18. Added: 3.54 Ultrasonic Shear Wave definition. | 45. Revised: 13.1. |
| 19. Added: New paragraphs 5.1.1 and 5.1.2. | 46. Revised: 14.1. |
| 20. Revised: 5.2.1.1. | 47. Added: Paragraph 14.2. |
| 21. Revised: 5.2.1.2. | 48. Revised: Recommended Minimum Training and Experience Levels – Tables 1–7. All tables revised to include experience in months and hours. |
| 22. Revised: 5.2.2.7. | 49. Deleted: Note 6. Original verbiage revised under Other. |
| 23. Deleted: 5.3.2.8 Standard and nonstandard analysis. | 50. Revised: Other. |
| 24. Added: 5.3.2.10 Prepare/review reports. | 51. Deleted: Appendix A – Inquiries. |
| 25. Added: 5.3.2.11 Review of customer specification requirements. | 52. Changed: Appendix B – Interpretation Policy to Appendix A. |
| 26. Deleted: 5.3.3.5 Final report analysis review. | 53. Revised: A.1. |
| 27. Revised: 6.3. | 54. Revised: A.2.6. |
| | 55. Deleted: ILI-PQ Inquiry Form. |
| | 56. Added: Summary of Changes in the 2024 Edition. |

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IN-LINE INSPECTION PERSONNEL QUALIFICATION AND CERTIFICATION

1.0 Scope

- 1.1 This Standard has been developed by The American Society for Nondestructive Testing Inc., to establish minimum requirements for the qualification and certification of in-line inspection (ILI) personnel whose jobs require specific knowledge of the technical principles of ILI technologies, operations, regulatory requirements, and industry standards as applicable to pipeline systems.
- 1.2 Qualification and certification of personnel shall be the responsibility of the employer.

2.0 Applicable References, Codes, Standards, Specifications, and Regulations

- 2.1 ASNT, current edition, Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing*, American Society for Nondestructive Testing, Columbus, OH*
- 2.2 ASNT, current edition, ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification and Certification of Nondestructive Testing Personnel*, American Society for Nondestructive Testing, Columbus, OH*
- 2.3 ASNT, current edition, *Interpreting SNT-TC-1A*, American Society for Nondestructive Testing, Columbus, OH*
- 2.4 API, 2021, API Std 1163: *In-line Inspection Systems Qualification Standard*, Third Edition, American Petroleum Institute, Washington, DC
- 2.5 API, 2013, API Std 1160, *Managing System Integrity for Hazardous Liquid Pipelines*, second edition, American Petroleum Institute, Washington, DC
- 2.6 ASME, 2014, ASME B31.8S-2014, *Managing System Integrity of Gas Pipelines*, ASME International, New York, NY
- 2.7 NACE, 2010, NACE Standard SP0102-2010, *Standard Recommended Practice, In-Line Inspection of Pipelines*, NACE International, Houston, TX
- 2.8 NACE, 2012, NACE Publication 35100, *In-Line Inspection of Pipelines*, NACE International, Houston, TX
- 2.9 Rijswijk, B.V., 2009, *Specification and requirements for intelligent pig inspection of pipelines – Pipeline Operators Forum (POF)*, Shell International Exploration and Production, The Netherlands
- 2.10 USDOT 49 CFR Parts 190–199, U.S. Government Printing Office, revised 2016

3.0 Definitions

- 3.1 **Aboveground Marker (AGM):** a portable device placed on the ground above a pipeline that both detects and records the passage of an in-line inspection tool or transmits a signal that is detected and recorded by the tool.
- 3.2 **Advanced Data Analysis:** evaluation of data not routinely performed during the analysis of routine datasets.
- 3.3 **Advanced Data Analyst:** a person responsible for the analysis of data beyond that which is routinely performed and supported by the ILI analysis equipment.
- 3.4 **AGM Identification:** the process of recognizing and understanding AGM indications within a dataset.
- 3.5 **AGM Mover and Tool Tracker:** any person who supervises or oversees the placement of AGMs. Also, those people who track an inspection tool for the purpose of maintaining point-to-point positioning of the inspection tool.
- 3.6 **Anomaly:** possible unexplained deviation from the norm in sound pipe material, materials, or welds.
- 3.7 **Anomaly Interaction Rules:** criteria that defines the interrelationship between anomalies.
- 3.8 **Appurtenance:** a component that is attached to the pipeline (e.g., valve, tee, casing, instrument connection).
- 3.9 **Axial Magnetic Flux Technology:** an ILI tool that magnetizes the pipe wall parallel to the axis of the pipe with sensors detecting flux leakage oriented in axial, radial, or circumferential direction, recently in two or all three directions.

*References available from the American Society for Nondestructive Testing (ASNT).

- 3.10 **Certification:** documented testimony of qualification.
- 3.11 **Certifying Agency:** the employer (or the employer's agent) of personnel being certified with the ultimate responsibility for certification resting with the employer.
- 3.12 **Certifying Authority:** the person or persons properly designated in the written practice to sign certifications on behalf of the employer.
- 3.13 **Closed-Book Examination:** an examination administered without access to reference material except for materials supplied with or in the examination.
- 3.14 **Comparable:** being at an equivalent or similar level of ILI responsibility and difficulty as determined by the employer.
- 3.15 **Component:** any physical part of the pipeline, other than line pipe, including but not limited to valves, welds, tees, flanges, fittings, taps, branch connections, outlets, supports, and anchors.
- 3.16 **Covered Task:** any job duty that this Standard deems to be critical in the execution of an in-line inspection and, therefore, requires qualification.
- 3.17 **Customer Specifications Reviewer:** the person who reviews contract feature specifications and verifies that the final report meets those specifications.
- 3.18 **Data Analysis:** the process through which indications recorded in an ILI are evaluated to classify, characterize, and size them.
- 3.19 **Data Analyst:** any person performing data-analysis covered tasks.
- 3.21 **Data Preparer/Processor:** any person who performs data processing or data preparation.
- 3.20 **Data Processing/Preparation:** the preparation or processing of the inspection data prior to data analysis.
- 3.22 **Electromagnetic Acoustic Transducer (EMAT):** a transducer for noncontact sound generation and reception directly generated within the material adjacent to the transducer.
- 3.23 **Employer:** the corporate, private, public entity, or organization, which employs personnel directly or indirectly for wages, salary, fees, or other considerations for the performance of ILI activities. This would include employers who obtain their qualified supplemental workforce personnel through third-party agencies, providing the use and certification of those supplemental employees is addressed in the employer's written practice.
- 3.24 **Experience:** work activities accomplished in a specific ILI technology under the direction of qualified Level II or III supervision with experience in that specific area, including the performance of the ILI technology and related activities, but not including time spent in organized training programs.
- 3.25 **Feature:** any physical object detected by an ILI tool during the performance of an inspection run. Note: Features may be anomalies, components, nearby metallic objects, welds, appurtenances, or some other item.
- 3.26 **Feature Categorization and Evaluation:** the process of identifying the type of pipeline features observed and determining their condition.
- 3.27 **Feature Detection and Location:**
 - 3.27.1 **Tool Operator:** field reviews of inspection data for the purpose of excavations or identifying tie-in sites for multiple inspection passes.
 - 3.27.2 **Data Analyst:** using in-line inspection data for the detection of pipeline features and the positioning of those features upon the pipeline for reporting.
- 3.28 **Feature Sizing:** the dimensioning of the observed feature (e.g., length, width, depth, etc.).
- 3.29 **Final Report:** a report that provides comprehensive analysis of the data from an in-line inspection.
- 3.30 **Final Report Analysis Review:** the technical review of the final report against contractual requirements including the final review of the analysis work.
- 3.31 **Final Report Review:** the commercial review of the final report against contractual requirements.
- 3.32 **In-Field Inspection Data Confirmation:** the initial review of inspection tool data in the field for the purpose of confirming the quality of the data.
- 3.33 **In-Line Inspection (ILI):** the inspection of a pipeline from the interior of the pipe using an in-line instrumented inspection tool. Also called *intelligent* or *smart pigging*.
- 3.34 **In-Line Inspection Technology:** describes the methodology (e.g., geometry, magnetic flux leakage, EMAT, ultrasonic testing, mapping, etc.) used in the performance of in-line inspection.
- 3.35 **In-Line Inspection Tool:** any device or vehicle that uses a nondestructive testing technique to collect data on the pipeline from the interior.
- 3.36 **Inspection Tool Maintenance:** the work performed on the inspection tool in the course of normal maintenance.
- 3.37 **Inspection Tool Preparation and Verification:** assembly, configuration, and confirmation of the inspection tool to a defined build state.

- 3.38 **Inspection Tool Repair:** the work on the inspection tool for the purpose of refurbishment to a given state.
- 3.39 **Integration and Confirmation of Supplementary Data:** the collating of multiple datasets and confirming that they are valid and appropriate for data analysis.
- 3.40 **Mapping Pig:** a configuration pig that uses inertial sensing or some other technology to collect data that can be analyzed to produce an elevation and plan view of the pipeline route. (Additionally, the operation of mapping tools is based on inertial navigation using built-in gyroscopes and accelerometers. The data acquired are X, Y, Z angular change and X, Y, Z velocity change.)
- 3.41 **Oversee Inspection Tool Launch, Run, and Receive:** ensuring the proper handling of the inspection tool during setup, loading, launching, and retrieving. Also being responsible for coordinating the AGM movement and tool tracking.
- 3.42 **Pipeline:** all parts of a facility through which a liquid or gas moves in transportation, including, but not limited to, line pipe, valves, and other appurtenances connected to line pipe, pumping units, fabricated assemblies associated with compressor units, metering and delivery stations and fabricated assemblies therein, and breakout tanks.
- 3.43 **Project Specific or Nonstandard Analysis:** the management and analysis of data resulting from a tool run in a nonstandard build state or unexpected run, or tool conditions such as tool speed outside limits, tool damage, and line product conditions.
- 3.44 **Project Specific or Nonstandard ILI Tool Modifications:** modifications to the inspection tool above what would be considered standard for the purpose of meeting a specific project or line requirement.
- 3.45 **Qualification:** demonstrated skill and knowledge, along with documented education, training, and experience required for personnel to properly perform the duties of a specific task.
- 3.46 **Qualified Individual:** an individual who can perform assigned covered tasks according to the employer's written practice.
- 3.47 **Qualifying Agency:** person, persons, or entity that provides services to qualify individuals against specific criteria within the employer's written practice.
- 3.48 **Report Generation:** production of a report (electronic or hard copy) apart from any data analysis.
- 3.49 **Site Assessment Survey:** a review of a pipeline and the operator's run plan for determining the compatibility of that pipeline with in-line inspection requirements.
- 3.50 **Training:** an organized program developed to impart the knowledge and skills necessary for qualification.
- 3.51 **Transverse Magnetic Field Technology:** circumferential (transverse) magnetization, perpendicular to the axis of the pipe to remove the interference on tool performance.
- 3.52 **Tool Operator:** any person performing a Tool Operator-covered task.
- 3.53 **Ultrasonic Compression Wave Technology:** an ILI tool that generates a normal compression beam through the pipe wall.
- 3.54 **Ultrasonic Shear Wave:** pertaining to pipe inspection, shear waves are generated in the pipe wall by transmitting ultrasonic pulses through a liquid medium or using EMAT technology without a liquid couplant.
- 3.55 **Verify:** establish that the task or requirement is completed as intended.
- 3.56 **Written Practice:** a written procedure developed by the employer that details the requirements for qualification and certification of its employees.

4.0 Written Practice

- 4.1 Employer shall establish a written practice for the control and administration of ILI personnel training, examination, and certification.
- 4.2 The employer's written practice shall describe the responsibility of each level of qualification.
- 4.3 The employer's written practice shall describe the training, experience, examination, and continuation requirements for each level of qualification.
- 4.4 The employer's written practice shall be reviewed and approved by the employer's designated management personnel every three (3) years.
- 4.5 The employer's written practice shall be maintained on file. Anywhere the phrase "written practice" appears means that the procedure is established, documented, implemented, and maintained.

5.0 Levels of Qualification

- 5.1 This standard defines three (3) levels of qualification for ILI personnel: Level I, Level II, and Level III. An individual in the process of becoming qualified to ILI Level I shall be considered a *trainee*. A trainee shall not independently conduct tests or interpret, evaluate, or report test results of any in-line inspection. A trainee shall perform all covered tasks under the supervision of Level II or III individuals qualified for the task performed.
 - 5.1.1 Emerging technologies will require the Level III of the Certifying Agency to establish criteria for assessing comparable qualifications between ILI technologies and other certification levels as described in the written practice.
 - 5.1.2 A prerequisite for qualification and certification in an emerging technology would require a minimum Level II qualification in the core technology (e.g., magnetic flux leakage and geometry). The written practice shall establish the criteria for certification in the emerging technology at or above a comparable technology requirement.
- 5.2 ILI Tool Operator Qualifications
 - 5.2.1 ILI Tool Operator Level I: shall be qualified to perform, as a minimum, the following covered tasks:
 - 5.2.1.1 ILI tool basic operation as described in the written practice.
 - 5.2.1.2 ILI tool maintenance and/or repair.
 - 5.2.1.3 Provide operational training in the use of AGMs.
 - 5.2.2 ILI Tool Operator Level II: in addition to the Level I tasks, an ILI Tool Operator Level II shall be qualified, as a minimum, to perform or verify the following covered tasks:
 - 5.2.2.1 Verification of job requirements.
 - 5.2.2.2 ILI tool preparations and build-state verifications.
 - 5.2.2.3 Project-specific ILI tool modifications.
 - 5.2.2.4 Oversee ILI tool launches, runs, and receiving.
 - 5.2.2.5 In-field inspection run data confirmations.
 - 5.2.2.6 Integration and confirmation of supplementary data.
 - 5.2.2.7 Feature detection and location as defined in paragraph 3.27.
 - 5.2.2.8 Field report.
 - 5.2.2.9 Provide supervision, training, and guidance to ILI Tool Operator trainees and Level I personnel.
 - 5.2.2.10 Review the customer's ILI standard operational and communications procedures (SOPs) to verify that each inspection can be conducted safely.
 - 5.2.3 ILI Tool Operator Level III: in addition to the Level II tasks, an ILI Tool Operator Level III individual shall be qualified, as a minimum, to perform or verify the following covered tasks:
 - 5.2.3.1 Establish techniques and procedures.
 - 5.2.3.2 Interpret codes and standards.
 - 5.2.3.3 Provide supervision, training, and guidance to ILI Tool Operator trainees, Level I, Level II, and other Level III personnel.
- 5.3 ILI Data Analyst Qualifications
 - 5.3.1 ILI Data Analyst Level I: an ILI Data Analyst Level I individual shall be qualified to perform, as a minimum, the following covered tasks:
 - 5.3.1.1 Basic analysis as described in employer's written practice.
 - 5.3.1.2 Review of customer specification requirements.
 - 5.3.2 ILI Data Analyst Level II: in addition to the Level I tasks, an ILI Data Analyst Level II individual shall be qualified, as a minimum, to perform or verify the following covered tasks (where applicable, based upon tool type):
 - 5.3.2.1 In-field inspection data confirmation.
 - 5.3.2.2 Integration and confirmation of supplementary data.
 - 5.3.2.3 Data preparation and processing.
 - 5.3.2.4 Feature detection and location as defined in paragraph 3.27.
 - 5.3.2.5 Feature categorization and evaluation.
 - 5.3.2.6 Feature sizing.
 - 5.3.2.7 Application of anomaly interaction rules.
 - 5.3.2.8 Organize and report ILI tool results.
 - 5.3.2.9 Provide supervision, training, and guidance to ILI Data Analyst trainees and Level I personnel.
 - 5.3.2.10 Prepare/review reports.
 - 5.3.2.11 Review of customer specification requirements.

- 5.3.3 ILI Data Analyst Level III: in addition to the Level II tasks, an ILI Data Analyst Level III individual shall be qualified, as a minimum, to perform or verify the following covered tasks:
 - 5.3.3.1 Establish techniques and procedures.
 - 5.3.3.2 Interpret codes and standards.
 - 5.3.3.3 Review of customer specification requirements.
 - 5.3.3.4 Advanced data analysis.
 - 5.3.3.5 Provide supervision, training, and guidance to ILI Data Analyst trainees, Level I, Level II, and other Level III personnel.

6.0 Education, Training, and Experience Requirements

- 6.1 Candidates for ILI qualification shall have sufficient education, training, and experience to verify qualification in those ILI technologies in which they are being considered for certification. Documentation of prior certification may be used by an employer as evidence of qualification for comparable levels of certification.
- 6.2 Documented training and/or experience in positions and activities comparable to those of ILI Levels I, II, or III prior to the establishment of the employer's written practice may be considered as satisfying the qualification requirements.
- 6.3 Tables 1A–4B list the recommended minimum training and experience factors to be considered by the employer in establishing a written practice for qualification of ILI Level I, II, and III personnel. The factors within these tables include the following criteria:
 - 6.3.1 Experience is cumulative within the specified ILI technology.
 - 6.3.2 Training hours are cumulative within the specified ILI technology.
 - 6.3.3 The employer's written practice shall include elements of these tables as applicable.
- 6.4 The training hours and experience months may be adjusted from Tables 1A–4B as described in the employer's written practice depending on the candidate's actual education level (e.g., grammar school, college graduate in engineering, etc.) and previous experience, as approved by the Level III.
- 6.5 A data analyst shall be a high school graduate.
- 6.6 Experience may be shared between ILI technologies provided those technologies target the same or similar feature types, as described in the employer's written practice.
- 6.7 In developing training, experience, and education requirements for new ILI technologies (e.g., next generation, multiple ILI tool, etc.), the employer shall review the detailed recommendations presented herein and modify them, as necessary, to meet particular needs.
 - 6.7.1 ILI Level III: the recommended minimum training and experience requirements shall be defined in the employer's written practice.

7.0 Training Programs

- 7.1 Personnel being considered for qualification to any level shall complete sufficient organized training to become thoroughly familiar with the principles and practices of the specified ILI technology related to the level of certification desired and applicable to the processes and equipment to be used.
- 7.2 The training program shall include sufficiently documented examinations to verify understanding of the necessary information.
- 7.3 The employer who utilizes outside training services is responsible for verifying that such services meet the requirements of the employer's written practice.

8.0 Examinations

- 8.1 Administration and Grading
 - 8.1.1 The employer shall be responsible for designating an ILI Level III for the administration and grading of examinations specified in the written practice. It shall be the responsibility of the ILI Level III, in the technologies in which certified, for the administration and grading of examinations specified in the written practice for ILI Level I, II, or other Level III personnel. The administration and grading of the examinations may be delegated to a qualified representative of the ILI Level III and so recorded. The actual administration and grading of ILI Level III examinations, as specified in the written practice, may be performed by a qualified representative of the employer.
 - 8.1.2 A written, electronic, or oral ten (10) question minimum General examination and a practical examination shall be administered to assess the qualifications of personnel. Administered exams and results shall be documented and retained on file for at least 12 months beyond the length of employment of the individual, along with any additional written comments.

- 8.1.3 Exam grades may be given either a numerical value or a pass/fail notation. The employer's written practice shall include such criteria for each exam.
- 8.1.4 The employer who utilizes outside examination services is responsible for verifying that the services meet the requirements of the employer's written practice.
- 8.2 Vision Examinations
 - 8.2.1 Near-Vision Acuity: where near-vision acuity is critical to job performance, certified ILI personnel shall be given an annual vision examination to verify natural or corrected near-distance acuity exists in at least one (1) eye such that the individual is capable of reading a Jaeger No. 2 or equivalent type chart and size letters at a distance of not less than 12 in. (30.5 cm) on a standard Jaeger test chart. The ability to perceive an Ortho-Rater minimum of 8 or similar test pattern is also acceptable.
 - 8.2.2 Color-Contrast Differentiation: where differentiation of colors is critical to job performance, the capability to distinguish and differentiate contrast among colors or shades of gray used in the ILI technology shall be demonstrated. This examination shall be administered prior to initial certification and thereafter as described in the employer's written practice.
- 8.3 Examination Requirements
 - 8.3.1 ILI Tool Operator Examinations
 - 8.3.1.1 Level I examinations shall address the basic principles of the applicable tasks listed in paragraph 5.2.1 and shall, at a minimum, consist of written/electronic and practical components.
 - 8.3.1.2 Level II examinations shall address the principles listed in paragraphs 5.2.1 and 5.2.2 and shall, at a minimum, consist of written/electronic and practical components.
 - 8.3.1.3 Level III examinations shall address the principles listed in paragraph 5.2, and shall consist of written/electronic, oral, and practical components.
 - 8.3.2 ILI Data Analyst Examinations
 - 8.3.2.1 Level I examinations shall address the basic principles of the applicable tasks listed in paragraph 5.3.1 and shall, at a minimum, consist of written/electronic and practical components.
 - 8.3.2.2 Level II examinations shall address the principles listed in paragraphs 5.3.1 and 5.3.2 and shall at a minimum consist of written/electronic and practical components.
 - 8.3.2.3 Level III examinations shall address the principles listed in paragraph 5.3 and shall consist of written/electronic, oral, and practical components.
 - 8.3.3 Examinations (written, electronic, or oral) shall be closed book. Questions utilizing provided reference materials shall require an understanding of the information rather than merely locating the appropriate answer. All questions used for examinations shall be approved by the employer's qualifying agent. For oral examinations, the examiner shall be responsible for documenting the individual's responses to questions asked. When preparing the examinations, the examiner shall select or devise appropriate questions to assess the ILI personnel's understanding and knowledge of the employer's procedures to complete applicable tasks, as well as the ILI personnel's ability to perform the task.
 - 8.3.4 Practical examinations shall be "hands-on" examinations that demonstrate the candidate's proficiency to perform covered task(s). The performance shall be measured based on at least ten (10) performance checkpoints selected to demonstrate an understanding of test variables and procedural requirements. The examiner shall be responsible for documenting the individual's responses and performance for each of the checkpoints.
 - 8.3.5 All examination questions missed, or practical performance not adequately demonstrated, shall be reviewed with the examinee to verify immediate and specific feedback for learning and documented.
- 8.4 Reexamination
 - 8.4.1 Those failing to attain the passing grades as determined by the employer's written practice shall wait at least 30 days or receive suitable additional training as determined by the employer before reexamination.

9.0 Certification

- 9.1 Certification shall be based upon the satisfactory completion of the following qualification requirements as defined in the employer's written practice:
 - 9.1.1 Education,
 - 9.1.2 Training,
 - 9.1.3 Experience,
 - 9.1.4 Examination.

- 9.2 Certification is technology-specific and may be further subdivided to address specific feature types, as defined in the employer's written practice.
- 9.3 The Certifying Authority, designated by the employer, shall verify and document that the personnel meet the specified qualification requirements outlined in paragraph 9.1.
- 9.4 When outside agencies supply certification services, the responsibility for ensuring that these organizations or individuals satisfy the requirements of this written practice on a continuing basis remains with the employer.
- 9.5 If outside agencies are used, the responsibility for certification shall be retained by the employer.
- 9.6 Personnel certification records shall be maintained on file by the employer for the duration specified in the employer's written practice and shall include the following:
 - 9.6.1 Name of certified ILI personnel.
 - 9.6.2 Level of certification and ILI technology.
 - 9.6.3 Educational background and qualified experience of certified ILI personnel.
 - 9.6.4 Statement indicating satisfactory completion of training in accordance with the employer's written practice.
 - 9.6.5 Results of the vision examinations prescribed in paragraph 8.2 for the current certification period.
 - 9.6.6 Current examination copies or evidence of successful completion of examinations.
 - 9.6.7 Other suitable evidence of satisfactory qualifications when such qualifications are used in lieu of the specific examination prescribed in the employer's written practice.
 - 9.6.8 Evidence of grades attained.
 - 9.6.9 Signature of Certifying Authority.
 - 9.6.10 Date of certification.
 - 9.6.11 Expiration date of certification.

10.0 Technical Performance Evaluation

- 10.1 At the discretion of the employer, any ILI personnel may be reexamined at any time during the stated period of certification in any or all of the examination parts and have their certifications extended or revoked. Supplemental training may also be required.
- 10.2 Periodically, as defined in the employer's written practice, the technical performance of certified ILI personnel shall be evaluated and documented by the employer. This evaluation and documentation shall follow the format and requirements described in paragraph 8.3 (i.e., practical, oral, and/or written exams).

11.0 Interrupted Service

- 11.1 The employer's written practice shall address interrupted service, which exceeds a period greater than six (6) months (i.e., in the event that the employee is reassigned to another job where certified activity is not used).
- 11.2 The written practice shall specify the evaluation requirements or the need for reexamination and/or recertification for the interrupted service.

12.0 Recertification

- 12.1 Certified ILI personnel shall be recertified every three (3) years based upon supplemental training as determined by the ILI Level III and one of the following:
 - 12.1.1 Evidence of continuing satisfactory technical performance as evaluated and documented by the ILI Level III; or
 - 12.1.2 Reexamination in any or all parts of the examinations for the applicable ILI technology(ies) as deemed necessary by the ILI Level III.

13.0 Termination of Certification

- 13.1 Certification(s) shall be revoked if certified ILI personnel fail to requalify for certification or if employment is terminated.

14.0 Reinstatement of Certification

- 14.1 Previously certified ILI personnel whose certifications have been terminated may have them reinstated to their former level without further examination provided:
 - 14.1.1 The certification records have been maintained in accordance with the employer's written practice; and,
 - 14.1.2 The employee's certification did not expire during the period of termination; and,
 - 14.1.3 The employee's certification is reinstated within six (6) months of termination.
 - 14.1.4 The employee's certification was not terminated due to technical capability or misconduct issues.
- 14.2 Requirements for personnel whose absence exceeds six (6) months shall be as defined in the written practice.

RECOMMENDED MINIMUM TRAINING AND EXPERIENCE LEVELS - TABLES 1-7

Geometry Technology

TABLE 1A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	3	40	High school graduate or equivalent
Level II	12	60	High school graduate or equivalent
Level III	18	100	High school graduate or equivalent

TABLE 1B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	3	40	High school graduate or equivalent
Level II	12	60	High school graduate or equivalent
Level III	18	100	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Axial Magnetic Flux Technology

TABLE 2A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	6	80	High school graduate or equivalent
Level II	18	160	High school graduate or equivalent
Level III	30	320	High school graduate or equivalent

TABLE 2B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	6	80	High school graduate or equivalent
Level II	18	160	High school graduate or equivalent
Level III	36	500	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Transverse Magnetic Field Technology

TABLE 3A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	6	80	High school graduate or equivalent
Level II	18	160	High school graduate or equivalent
Level III	30	280	High school graduate or equivalent

TABLE 3B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	12	80	High school graduate or equivalent
Level II	24	160	High school graduate or equivalent
Level III	36	500	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Ultrasonic Compression Wave Technology

TABLE 4A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	9	80	High school graduate or equivalent
Level II	24	160	High school graduate or equivalent
Level III	36	280	High school graduate or equivalent

TABLE 4B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	12	80	High school graduate or equivalent
Level II	24	160	High school graduate or equivalent
Level III	60	500	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Ultrasonic Shear Wave Technology

TABLE 5A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	9	120	High school graduate or equivalent
Level II	24	200	High school graduate or equivalent
Level III	36	320	High school graduate or equivalent

TABLE 5B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	12	240	High school graduate or equivalent
Level II	24	400	High school graduate or equivalent
Level III	60	600	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

EMAT Technology

TABLE 6A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	9	120	High school graduate or equivalent
Level II	24	200	High school graduate or equivalent
Level III	36	320	High school graduate or equivalent

TABLE 6B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	12	240	High school graduate or equivalent
Level II	24	400	High school graduate or equivalent
Level III	60	600	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Mapping Technology

TABLE 7A
ILI Tool Operator

Level	Experience (months)	Training (hours)	Education (formal)
Level I	3	40	High school graduate or equivalent.
Level II	12	80	High school graduate or equivalent.
Level III	18	120	High school graduate or equivalent.

TABLE 7B
ILI Data Analyst

Level	Experience (months)	Training (hours)	Education (formal)
Level I	3	40	High school graduate or equivalent.
Level II	12	80	High school graduate or equivalent.
Level III	18	120	A degree from a university, college, or technical school. In lieu of a degree, an additional one (1) year of experience is required for mapping or geometry technologies and an additional two (2) years of experience is required for all other technologies.

Notes:

1. Experience and training are cumulative.
2. Training shall be outlined in employer’s written practice.
3. Experience with comparable ILI technologies can be considered as described in paragraph 6.6 and per employer’s written practice.
4. Emerging technologies require a Level I, Level II, or Level III ILI Tool Operator/ILI Data Analyst from a comparable ILI technology as per employer’s written practice.

APPENDIX

Appendix A – Interpretation Policy

(Not a part of American National Standard ANSI/ASNT ILI-PQ-2023.)

- A.1 Purpose. It is the responsibility of the ASNT Standards Council (StC) to issue interpretations of standards it has developed where questions regarding the proper way to comply with such standards have arisen. This appendix establishes the procedures of the StC and those inquiring shall use them to make such interpretations.
- A.2 Inquiries. The following guidelines for preparing an inquiry will expedite consideration and response to the inquirer.
 - A.2.1 Each inquiry should be limited to a single aspect of the standard or previous interpretation of the standard. Inquiries that concern multiple inquiries or unrelated subjects may be returned for further clarification.
 - A.2.2 The specific question to be answered by the StC should be distinctly set apart from the body of the letter and must be in condensed, precise, question format, omitting superfluous background information and, if possible, asked in such a manner that a “yes” or “no” (perhaps with a short explanation) would be an acceptable reply.
 - A.2.3 The inquiry should state the purpose of the question, which could be to obtain a clarification of the standard’s mandates or the intent of the canvass group as it developed the standard.
 - A.2.4 The inquiry should contain the needed information for the StC’s understanding of the question.
 - A.2.5 The inquiry should include references to the applicable paragraph(s) of the standard or previous interpretation of the standard.
 - A.2.6 Inquiries must be in writing. Request for Interpretation forms are available on the ASNT website (asnt.org).

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